18.3 Gradient fields and path-independent fields, part 1

Fundamental Theorem of Calculus for Line Integrals (FTCLI)

Suppose C is a piecewise smooth oriented path with starting point P and ending point Q. If f is a function whose gradient is continuous on C, then

$$\int_C \nabla f \cdot d\vec{r} = f(Q) - f(P).$$

→ Do problem 1.

A vector field $\vec{F}$ is called conservative if it is a gradient field, i.e., if there is a function f such that $\vec{F} = \nabla f$. In this case f is called a potential function of $\vec{F}$.

Warning: Potential functions are not unique, and not all vector fields are conservative!

→ Do problem 2.